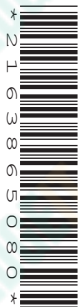


[illegible]

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9702/52

February/March 2019

1 hour 15 minutes

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

- 1 A student is investigating the behaviour of a capacitor-resistor circuit as shown in Fig. 1.1.

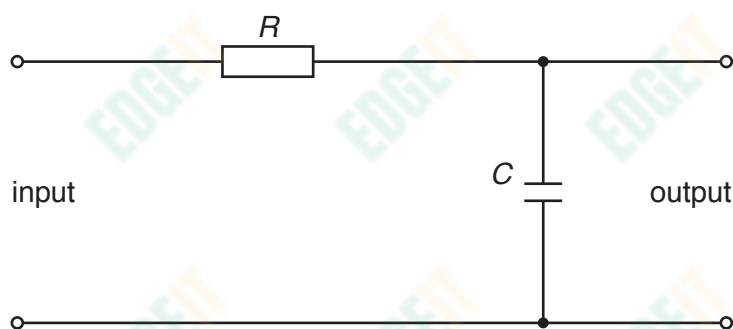


Fig. 1.1

A neon lamp flashes on and off when it is connected across the capacitor with a potential difference V_F across the lamp of approximately 90 V.

The student has a number of unmarked resistors.

It is suggested that the period T of the flashes of the lamp is related to the resistance R of the resistor by the expression

$$T = RCK$$

where C is the capacitance of the capacitor and K is a constant.

The constant K is given by

$$K = \ln \left(\frac{V_i - V_L}{V_i - V_F} \right)$$

where V_i is the potential difference across the input, V_F is the potential difference required to make the lamp flash and V_L is a constant.

Design a laboratory experiment to test the relationship between T and R .
Explain how your results could be used to determine a value for K and V_L .

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

This image shows a full page of primary-ruled paper. It features multiple sets of horizontal dashed lines spaced evenly down the page. Diagonal dotted lines cross these horizontal lines at approximately a 45-degree angle, creating a series of slanted rectangular boxes across the entire surface. The background is white, and the lines are light gray or blue. There is no handwriting or other markings on the page.

- 2 A student is investigating the motion of a small steel ball in cooking oil. A measure of the oil's resistance to the ball's motion is called viscosity. Viscosity has the units pascal second (Pa s).

The student drops a ball into a cylinder of oil as shown in Fig. 2.1.

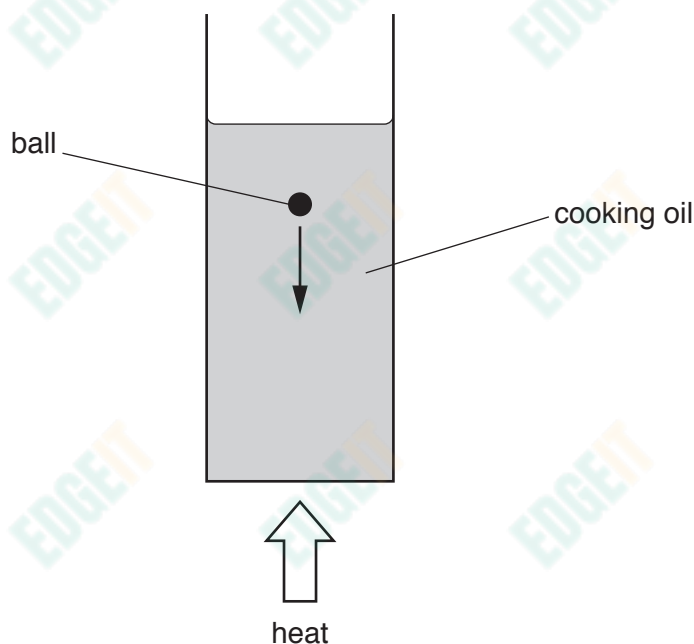


Fig. 2.1

The velocity of the ball is measured when it becomes constant and then the viscosity of the oil is determined.

The experiment is repeated for different temperatures of oil.

It is suggested that the viscosity η and the Celsius temperature θ are related by the equation

$$\eta = p\theta^q$$

where p and q are constants.

- (a) A graph is plotted of $\lg \eta$ on the y -axis against $\lg \theta$ on the x -axis.

Determine expressions for the gradient and the y -intercept.

gradient =

y -intercept =

[1]

(b) Values of θ and η are given in Fig. 2.2.

$\theta/^{\circ}\text{C}$	$\eta/10^{-3}\text{ Pa s}$	$\lg (\theta/^{\circ}\text{C})$	$\lg (\eta/10^{-3}\text{ Pa s})$
38	41 ± 1		
46	32 ± 1		
55	25 ± 1		
64	20 ± 1		
72	17 ± 1		
79	14 ± 1		

Fig. 2.2

Calculate and record values of $\lg (\theta/^{\circ}\text{C})$ and $\lg (\eta/10^{-3}\text{ Pa s})$ in Fig. 2.2.
Include the absolute uncertainties in $\lg (\eta/10^{-3}\text{ Pa s})$.

[2]

(c) (i) Plot a graph of $\lg (\eta/10^{-3}\text{ Pa s})$ against $\lg (\theta/^{\circ}\text{C})$.
Include error bars for $\lg (\eta/10^{-3}\text{ Pa s})$.

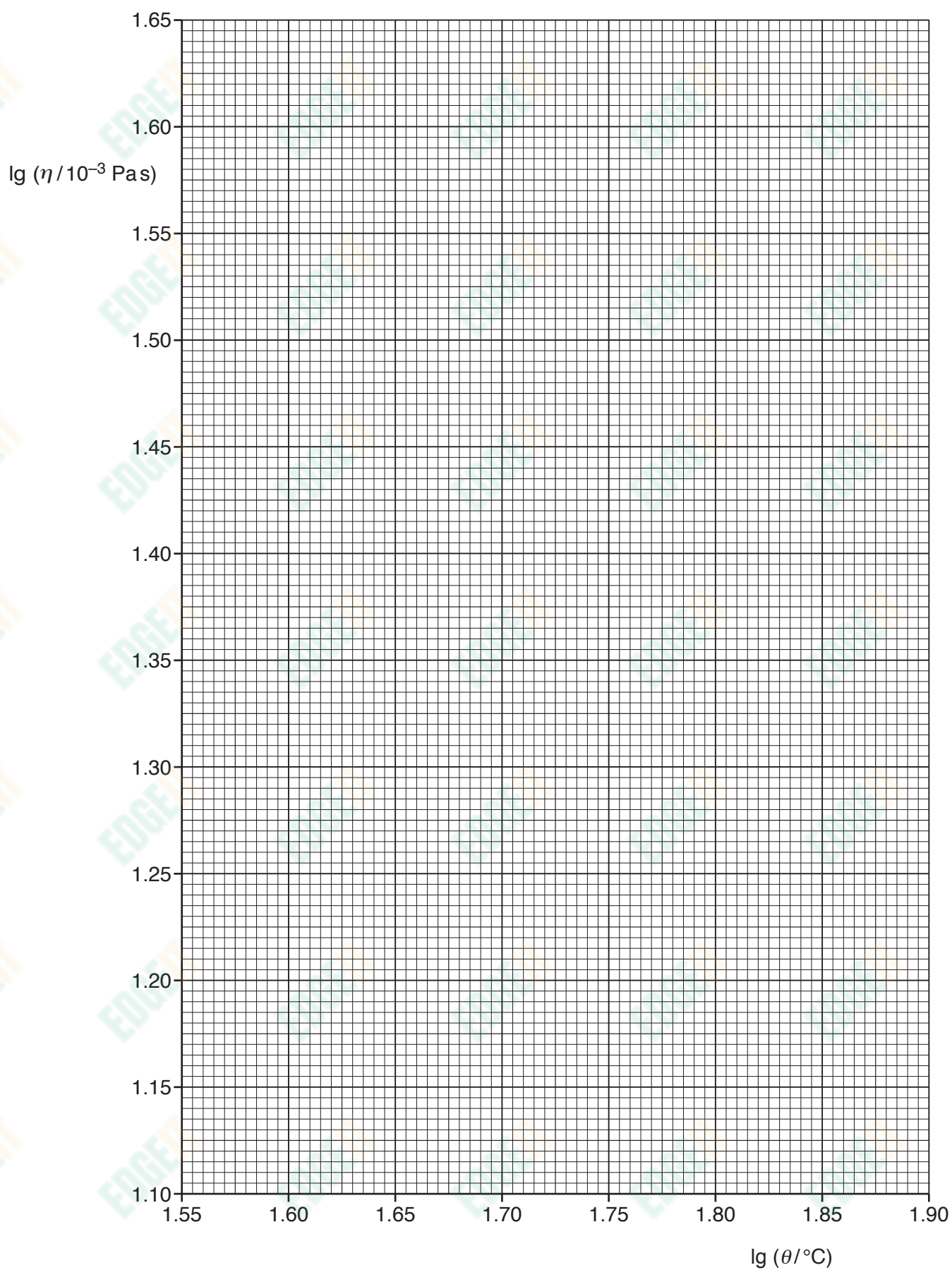
[2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Both lines should be clearly labelled.

[2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of p and q . Include the absolute uncertainties in your answer.

p =

q = [3]

- (e) Using your answers to (d), determine the temperature θ at which the viscosity of the cooking oil is 0.10 Pa.s.

θ = °C [1]

[Total: 15]

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